Imperial College London
AE3-304 Aircraft Structures – Plate Bending & Buckling

Tutorial 3

Questions

Consider a rectangular plate with clamped unloaded edges of length a and simply supported loaded edges of width b ($b < a$). A uniform compressive membrane force N_x acts on the loaded edges. The approximated transversal deformation of the buckled plate is described by the following family of shape functions (i.e. buckling modes)

$$w = A_{mn} \sin\frac{m\pi x}{a}\left(1 - \cos\frac{2n\pi y}{b}\right); \quad m, n \in \mathbb{N}^+$$

1) Explain why the shape function given above is suitable for predicting the buckling of the plate via the Rayleigh-Ritz method.

[10 marks]

2) Calculate the strain energy of the plate for the load and assumed deflection stated above.

[30 marks]

3) Calculate the external work performed by the in-plane membrane force for N_x the buckling modes given above.

[20 marks]

4) Find the expression of the buckling load of the plate as a function of the aspect ratio a/b.

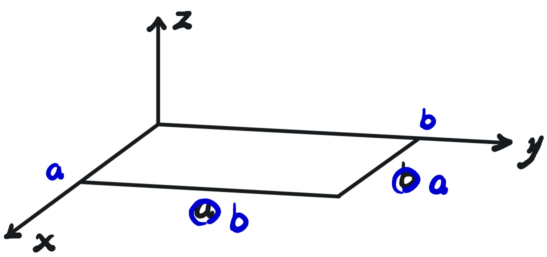
[20 marks]

5) Compute the aspect ratio of the plate a/b that yields the minimum buckling load, as well as the value of the latter.

[20 marks]

Some useful integrals:

$$\int_0^a \sin^2\frac{m\pi x}{a}dx = \int_0^a \cos^2\frac{m\pi x}{a}dx = \frac{a}{2}, \quad m \in \mathbb{N}^+$$



1) for Rayleigh-Ritz Method, essential (geometry) boundary condition should be satisfied

$\therefore w(x,0) = w(x,b) = 0$ for $0 \le x \le a$

$w(0,y) = w(a,y) = 0$ for $0 \le y \le b$

and $\boxed{\frac{dw}{dx}(x,0) = \frac{dw}{dx}(x,b) =} \frac{dw}{dy}(x,0) = \frac{dw}{dy}(x,b) = 0$ for $0 \le x \le a$

$\because w = A_{mn} \sin\frac{m\pi x}{a}\left(1 - \cos\frac{2n\pi y}{b}\right)$

$\therefore \frac{dw}{dx} = A_{mn} \frac{m\pi}{a} \cos\frac{m\pi x}{a}\left(1 - \cos\frac{2n\pi y}{b}\right)$

$\frac{dw}{dy} = A_{mn} \frac{2n\pi}{b} \sin\frac{m\pi x}{a} \sin\frac{2n\pi y}{b}$

$\therefore$ satisfied boundary condition listed above

$\frac{dw}{dx}(0,y) \neq 0$, $\frac{dw}{dx}(a,y) \neq 0$ for $0 \le y \le b$

* $\frac{dw}{dy}$ rotation along the edge parallel to the x-axis

$\frac{dw}{dx}$ rotation along the edge parallel to the y-axis

2) $U = \frac{1}{2}\iiint \sigma\varepsilon \, dV$

$$= \frac{1}{2}\int_0^a\int_0^b \int \sigma_{xx} z \, dz \frac{\varepsilon_{xx}}{z} + \int \sigma_{yy} z \, dz \frac{\varepsilon_{yy}}{z} + \int \tau_{xy} z \, dz \frac{\gamma_{xy}}{z} \, dx dy$$

$$= \frac{1}{2}\int_0^a\int_0^b M_x \frac{\varepsilon_{xx}}{z} + M_y \frac{\varepsilon_{yy}}{z} + M_{xy} \frac{\gamma_{xy}}{z} \, dx dy$$

$$= \frac{1}{2}D\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2}\right)\frac{\partial^2 w}{\partial x^2} + \left(\frac{\partial^2 w}{\partial y^2} + \nu\frac{\partial^2 w}{\partial x^2}\right)\frac{\partial^2 w}{\partial y^2} + (1-\nu)\frac{\partial^2 w}{\partial x \partial y}\left(2\frac{\partial^2 w}{\partial x \partial y}\right) dx dy$$

$$= \frac{1}{2}D\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x^2}\right)^2 + \left(\frac{\partial^2 w}{\partial y^2}\right)^2 + 2\nu\left(\frac{\partial^2 w}{\partial x^2}\frac{\partial^2 w}{\partial y^2}\right) + 2(1-\nu)\left(\frac{\partial^2 w}{\partial x \partial y}\right)^2 dx dy$$

where $D = \frac{Et^3}{12(1-\nu^2)}$

$\frac{\partial^2 w}{\partial x^2} = -A_{mn}\left(\frac{m\pi}{a}\right)^2 \sin\frac{m\pi x}{a}\left(1 - \cos\frac{2n\pi y}{b}\right)$

$\frac{\partial^2 w}{\partial y^2} = A_{mn}\left(\frac{2n\pi}{b}\right)^2 \sin\frac{m\pi x}{a} \cos\frac{2n\pi y}{b}$

$\frac{\partial^2 w}{\partial x \partial y} = A_{mn}\left(\frac{m\pi}{a}\right)\left(\frac{2n\pi}{b}\right) \cos\frac{m\pi x}{a} \sin\frac{2n\pi y}{b}$

$\therefore \left(\frac{\partial^2 w}{\partial x^2}\right)^2 = A_{mn}^2\left(\frac{m\pi}{a}\right)^4 \sin^2\frac{m\pi x}{a}\left(1 - 2\cos\frac{2n\pi y}{b} + \cos^2\frac{2n\pi y}{b}\right)$

$$\left(\frac{\partial^2 w}{\partial y^2}\right)^2 = A_{mn}^2 \left(\frac{2n\pi}{b}\right)^4 \sin^2\frac{m\pi x}{a} \cos^2\frac{2n\pi y}{b}$$

$$\frac{\partial^2 w}{\partial x^2}\frac{\partial^2 w}{\partial y^2} = -A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \left(\frac{2n\pi}{b}\right)^2 \sin^2\frac{m\pi x}{a} \left(\cos\frac{2n\pi y}{b} - \cos^2\frac{2n\pi y}{b}\right)$$

$$\left(\frac{\partial^2 w}{\partial x \partial y}\right)^2 = A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \left(\frac{2n\pi}{b}\right)^2 \cos^2\frac{m\pi x}{a} \sin^2\frac{2n\pi y}{b}$$

$$\therefore \int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x^2}\right)^2 dx\,dy = A_{mn}^2 \left(\frac{m\pi}{a}\right)^4 \left(\frac{a}{2}\right)\left(b - 0 + \frac{b}{2}\right) = A_{mn}^2 \frac{3m^4\pi^4 b}{4a^3}$$

$$\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial y^2}\right)^2 dx\,dy = A_{mn}^2 \left(\frac{2n\pi}{b}\right)^4 \left(\frac{a}{2}\right)\left(\frac{b}{2}\right) = A_{mn}^2 \frac{4n^4\pi^4 a}{b^3}$$

$$\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x^2}\frac{\partial^2 w}{\partial y^2}\right) dx\,dy = -A_{mn}^2 \left(\frac{4m^2n^2\pi^4}{a^2b^2}\right)\left(\frac{a}{2}\right)\left(0 - \frac{b}{2}\right) = A_{mn}^2 \frac{m^2n^2\pi^4}{ab}$$

$$\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x \partial y}\right) dx\,dy = A_{mn}^2 \left(\frac{4m^2n^2\pi^4}{a^2b^2}\right)\left(\frac{a}{2}\right)\left(\frac{b}{2}\right) = A_{mn}^2 \frac{m^2n^2\pi^4}{ab}$$

$$\therefore U = \frac{D}{2}\left[A_{mn}^2 \frac{3m^4\pi^4 b}{4a^3} + A_{mn}^2 \frac{4n^4\pi^4 a}{b^3} + 2\nu A_{mn}^2 \frac{m^2n^2\pi^4}{ab} + 2(1-\nu) A_{mn}^2 \frac{m^2n^2\pi^4}{ab} \right]$$

$$= \frac{1}{8} A_{mn}^2 D\pi^4 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2n^2}{ab}\right)$$

3) $$W_{ip} = \frac{1}{2}\int_0^a\int_0^b N_x \left(\frac{dw}{dx}\right)^2 dx\,dy$$

$$= \frac{N_x}{2}\int_0^a\int_0^b A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \cos^2\frac{m\pi x}{a} \left(1 - 2\cos\frac{2n\pi y}{b} + \cos^2\frac{2n\pi y}{b}\right) dx\,dy$$

$$= \frac{N_x}{2} A_{mn}^2 \left(\frac{m\pi}{a}\right)^2 \left(\frac{a}{2}\right)\left(b - 0 + \frac{b}{2}\right)$$

$$= \frac{1}{8} N_x A_{mn}^2 \pi^2 \left(\frac{3m^2 b}{a}\right)$$

4) $$\Pi = U - W = U - (W_{op} + W_{ip})$$

$$= \frac{1}{8} A_{mn}^2 \pi^2 \left[D\pi^2 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2n^2}{ab}\right) - N_x \left(\frac{3m^2 b}{a}\right) \right]$$

$$\therefore \frac{d\Pi}{dA_{mn}} = \frac{1}{4} A_{mn} \pi^2 \left[D\pi^2 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2n^2}{ab}\right) - N_x \left(\frac{3m^2 b}{a}\right) \right]$$

At the buckling load, N_{xB}, $\frac{d\Pi}{dA_{mn}} = 0$

$$\therefore D\pi^2 \left(\frac{3m^4 b}{a^3} + \frac{16n^4 a}{b^3} + \frac{8m^2n^2}{ab}\right) = N_{xB}\left(\frac{3m^2 b}{a}\right)$$

$$\therefore N_{xB} = D\pi^2 \left(\frac{m^2}{a^2} + \frac{16n^4 a^2}{3m^2 b^4} + \frac{8n^2}{3b^2}\right) = \frac{D\pi^2}{b^2}\left[m^2 \left(\frac{a}{b}\right)^{-2} + \frac{16n^4}{3m^2}\left(\frac{a}{b}\right)^2 + \frac{8n^2}{3} \right]$$

5) $\dfrac{dN_{xB}}{d(a/b)} = \dfrac{D\pi^2}{b^2}\left[-2m^2\left(\dfrac{a}{b}\right)^{-3} + \dfrac{32n^4}{3m^2}\left(\dfrac{a}{b}\right)\right]$

To find minimum buckling load, $\dfrac{dN_{xB}}{d(a/b)} = 0$

$\therefore\ -2m^2\left(\dfrac{a}{b}\right)^{-3} + \dfrac{32n^4}{3m^2}\left(\dfrac{a}{b}\right) = 0$

for $\dfrac{a}{b} \neq 0$ and $\dfrac{a}{b} > 0$,

$-6m^4 + 32n^4\left(\dfrac{a}{b}\right)^4 = 0$

$\therefore\ \dfrac{a}{b} = \dfrac{3^{\frac{1}{4}}m}{2n}$

$$\therefore (N_{xB})_{min} = \frac{D\pi^2}{b^2}\left(m^2\frac{4n^2}{\sqrt{3}m^2} + \frac{16n^4}{3m^2}\frac{\sqrt{3}m^2}{4n^2} + \frac{8n^2}{3}\right)$$

$$= \frac{D\pi^2}{b^2}\left(\frac{4\sqrt{3}}{3}n^2 + \frac{4\sqrt{3}}{3}n^2 + \frac{8n^2}{3}\right)$$

$$= \frac{8+8\sqrt{3}}{3}\,\frac{D\pi^2 n^2}{b^2}$$